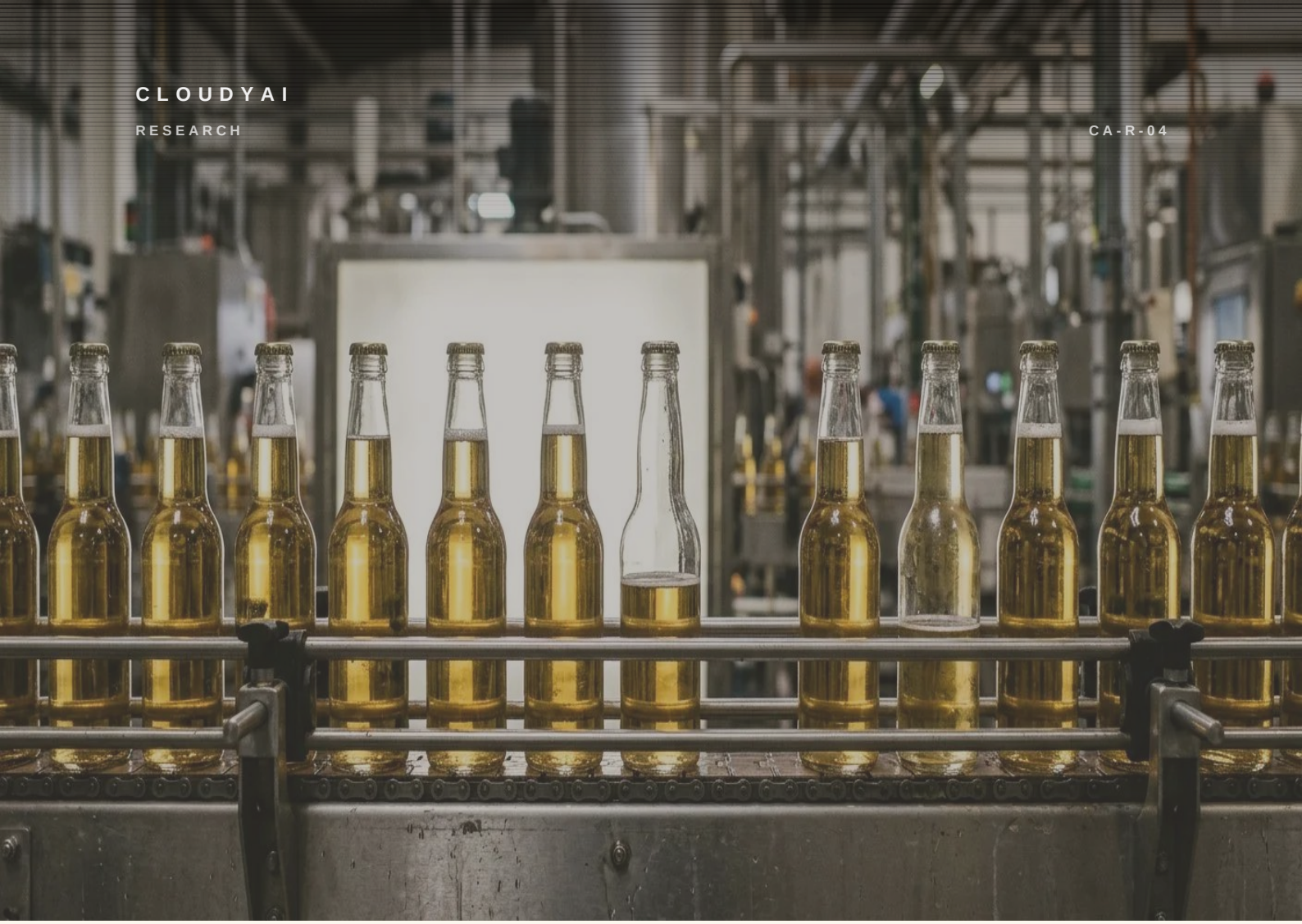


COMPUTER VISION

The Fill Line Is a Physics Problem

What backlit cameras can and cannot prove about fill level, and how to validate a system before the first reject.

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SUMMARY

Foam, meniscus, tinted glass, and opaque containers decide whether fill-level inspection works long before any model does. A practical note on optics, reject logic, and validating against a checkweigher.

Every bottling plant already knows whether its filler is drifting. The question is how quickly it finds out. Overfilling gives product away at a rate that is surprisingly easy to ignore; underfilling produces rejects, complaints, and in regulated categories a compliance problem. Both are measurable, and both are visible if you light the container correctly.

This paper is about the part that decides whether the project works: the optics and the reject logic. Model choice is the least interesting variable.

Why backlight is the default

Fill level is a boundary between liquid and gas. Backlighting turns that boundary into a high-contrast line: light passes through the liquid and the headspace differently, and a camera looking into a diffused panel sees the fill line as a dark or bright edge depending on the product. Front lighting mostly shows you the label and the reflection of the ceiling.

The best geometry is a camera square to the container, with the container filling 60 to 80 per cent of the frame height, and a diffused backlight panel immediately behind it. A 12 to 25 mm lens on a modern sensor is usually enough. The camera and the light must be rigid relative to the conveyor; a bracket that moves by a millimetre changes the measurement.

The four hard cases

Case	Why it defeats a naive system	Usual mitigation
Foam	Foam sits above the liquid line and reads as overfill	Settle time before inspection, or detect the liquid line beneath the foam with a trained model
Meniscus	The curve at the wall makes the edge ambiguous	Measure at a fixed x-offset from the container wall, or use the meniscus apex
Tinted or dark glass	Low contrast between liquid and headspace	Stronger backlight, longer exposure, or a wavelength the glass transmits
Opaque containers	Nothing transmits; the fill line is invisible	X-ray, checkweigher, or fill-height triangulation from the neck

Trigger, exposure, and the conveyor



Backlit fill inspection. Two underfilled containers in one frame - the case a manual spot check misses.

Do not run free. Trigger the camera from an encoder or a photo-eye so every container is imaged at the same point in the same pose. Use a short exposure with a strobe synchronised to the trigger; motion blur at 80,000 bottles per hour is measured in millimetres and it will look exactly like an ambiguous fill line. If the container is rotating or oscillating on the conveyor, add a mechanical guide before blaming the model.

From measurement to reject

The output of the vision system should be a number - estimated fill height in millimetres, plus a confidence - not a binary. The reject decision belongs to a small, auditable rules layer.

- Two thresholds, not one: a warning band and a reject band. Warning triggers a filler-head check; reject removes the container.
- Debounce across consecutive containers. A single outlier is usually a bubble, not a trend.
- Handshake with the rejector: the PLC needs to know which lane, which position, and whether the rejection was confirmed by a downstream sensor.
- Log every reject with the image, the measurement, the head number, and the timestamp. This is what turns a reject into a maintenance action.
- Never silently auto-adjust the filler. Recommend, do not command, until the system has earned trust.

False rejects cost more than they look

A reject at the filler is product, container, and capacity. A reject that is wrong also costs credibility: operators start bypassing the system, and once bypassed it is very hard to bring back. Before go-live, calculate the cost of a false reject on your line and set the operating threshold so that false rejects stay below a number the plant manager has agreed to in writing. It is usually better to miss a marginal underfill than to reject a good bottle every four minutes.

Validation against a checkweigher

The honest way to prove a fill-level system is to run it alongside an instrument the plant already trusts. Two shifts of side-by-side data is enough to see whether the system is measuring what you think it is measuring.

01 Run at normal line speed with production product. Do not validate on water and then run on beer.

02 Capture at least 2,000 containers with both systems, including deliberate underfills and overfills at the start and end of the run.

03 Compare agreement, bias, and the distribution of disagreement. A consistent bias is a calibration offset; a wide disagreement is a measurement problem.

04 Check performance at the extremes: coldest product, foamiest product, slowest line speed, fastest line speed.

05 Repeat after any change to lighting, lens, container supplier, or product.

> **The question to ask a vendor**
>
> Show me the distribution of measurement error on my product, at my line speed, across a full shift - not the accuracy on a demo video.

Limits

Backlit vision measures where the liquid appears to end. It does not measure mass, dissolved gas, or headspace pressure. For carbonated products, fill height at the filler is not the same as fill height after equilibration. For opaque containers you need a different sensor. And no camera fixes a filler that is genuinely drifting; it only tells you sooner.

ABOUT THIS PAPER

CloudyAI is a Zimbabwean AI studio. We deploy computer vision on existing plant cameras, audit AI spend, and help teams respond when an AI system misbehaves. Research is published free, without an email wall.

Corrections and questions: research@cloudyai.co.zw. If you would like this method applied to your plant, the first conversation is free and the camera audit is a fixed-price week.